

RealWear Industrial Operations Analytics

Smart Factory Intelligence System — End-to-End Analytics Report

Prepared By · Dev Pandey — Data Analyst

Project Period · January 2025 – January 2026

Dataset · 12,000 Sessions | 5 Plants | 20 Devices

Tools Used · Excel · SQL · Power BI · Python · Statistics

Document Type · Comprehensive Analytics Report

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1. Executive Summary

High-level findings and strategic outcomes

12,000

Total Sessions

77.50

Avg Productivity Score

71.98%

Issue Resolution Rate

9.95K

Downtime Saved (hrs)

92.91%

Session Completion Rate

Project Purpose

This report presents the findings of a comprehensive end-to-end analytics project conducted on operational data from RealWear Navigator 500 wearable devices deployed across five manufacturing plants in India. The analysis spans 12,000 remote assistance sessions recorded between January 2025 and January 2026, covering worker productivity, device health, platform performance, and voice command efficiency. The goal is to identify operational bottlenecks, validate business assumptions with statistical rigor, and deliver evidence-based recommendations to improve industrial remote assistance performance.

Critical Findings at a Glance

- MS Teams delivers 17% higher productivity than Webex with statistical certainty ($T = 120.04$). Standardization across all plants is immediately actionable.
- Voice command success rate ($r = 0.70$) and command failures ($r = -0.58$) are the only meaningful drivers of productivity. All other operational factors are negligible.
- Noise level significantly affects command failures ($p < 0.001$) but explains only 3.26% of total variance. It is not the primary driver of performance problems.
- External microphones appear to increase failures due to a confounding variable — noise level — not because they perform poorly. Current policy assumptions are flawed.
- Shift timing has zero meaningful effect on productivity ($\eta^2 = 0.0001$). Shift-based productivity interventions will produce no measurable return.
- 275 anomalous sessions with extreme command failures represent genuine operational incidents requiring dedicated investigation and root cause analysis.

2. Project Overview

Industry context, problem statement and objectives

Industry Context

Modern manufacturing plants increasingly rely on wearable technology to enable hands-free remote assistance on the shop floor. RealWear Navigator 500 devices allow field workers to receive expert guidance through voice-commanded sessions, dramatically reducing equipment downtime and accelerating issue resolution without requiring workers to disengage from their tasks.

The Problem

Despite widespread deployment across five Indian manufacturing plants — Chennai Refinery Unit, Jamshedpur Steel Plant, Bokaro Manufacturing Hub, Pune Automation Plant, and Vadodara Chemical Unit — management lacked a unified system to monitor operational efficiency. Workers across plants, departments, and shifts experienced varying levels of command failures, session drops, unresolved issues, and device degradation with no visibility into root causes or comparative benchmarks.

Business Objectives

1. Identify which plants and departments underperform and why
2. Determine whether noise level significantly affects voice command accuracy
3. Statistically compare MS Teams vs Webex performance across all metrics
4. Assess whether shift timing influences worker productivity
5. Evaluate the effectiveness of external microphones in noisy environments
6. Monitor device health and flag devices at risk of failure
7. Identify the strongest predictors of worker productivity
8. Deliver actionable business recommendations backed by statistical evidence

Technology Stack

Tool	Purpose	Deliverable
Microsoft Excel	Data cleaning, pivot analysis, feature engineering	Interactive dashboard + 8 pivot analyses
SQL (SQLite)	Advanced querying, window functions, CTEs	54 queries across 12 analytical categories
Python + scipy	Statistical testing, correlation, outlier detection	Hypothesis tests + correlation heatmap
Power BI	Interactive business intelligence dashboards	2 published dashboards
Pandas + NumPy	Data manipulation and computation	Statistical analysis notebook

3. Dataset Architecture

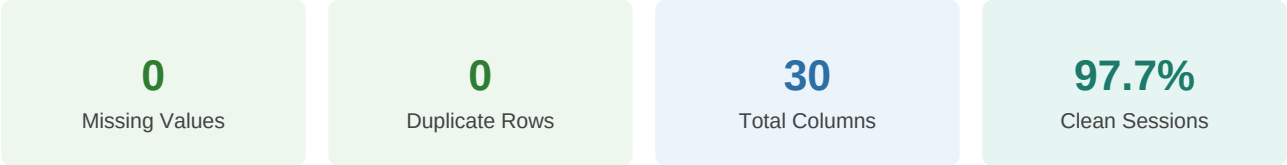
Schema design, relationships and data quality

Dataset Structure

The dataset follows a star schema with Master_Session_Log as the central fact table and five dimension tables. This structure mirrors enterprise-level industrial datasets and enables multi-dimensional analysis across workers, devices, plants, issues, and time.

Table	Type	Rows	Key Columns
Master_Session_Log	Fact	12,000	Session_ID, Worker_ID, Device_ID, Plant_Location, Issue_Type
Worker_Master	Dimension	7,000+	Worker_ID, Worker_Role, Department, Experience_Years
Device_Health_Log	Dimension	20	Device_ID, Health_Score, Battery_Level, Uptime_Hours
Plant_Department_Summary	Dimension	25	Plant_Location, Department, Avg_Productivity
Issue_Type_Analysis	Dimension	Multiple	Issue_Type, Avg_Resolution_Time, Resolution_Rate
Monthly_Trend	Dimension	13	Month_Year, Session_Count, Avg_Productivity
Data_Dictionary	Reference	30	Column_Name, Data_Type, Description

Data Quality Assessment



Data Quality

The dataset required no missing value imputation and no duplicate removal — an unusually clean industrial dataset. Only 2.29% of sessions (275) were flagged as anomalous through outlier detection, confirming high data integrity throughout.

4. Excel Analysis

Data cleaning, feature engineering, pivot analysis and dashboard

Approach

Excel was used as the primary exploratory analysis tool. Rather than modifying the raw dataset, a Working_Data sheet was created as a copy to preserve original data integrity. Three calculated columns were engineered to enable richer analysis.

Feature Engineering

Calculated Column	Formula Logic	Purpose
Command_Failure_Rate_%	(Command_Failures / Command_Attempts) x 100	Normalized failure metric per session
Resolution_Efficiency	Score based on resolution time vs maximum time	Captures both speed and success of resolution
Session_Efficiency_Score	(Productivity x 0.6) + (Resolution_Efficiency x 0.4) - (Command_Failure_Rate_%)	Composite performance score per session

Key Findings from Pivot Analysis

Plant Performance

Bokaro Manufacturing Hub leads with the highest average productivity score (78.04) while Vadodara Chemical Unit and Chennai Refinery Unit share the lowest position (77.15). The gap of 0.89 points is observable but requires statistical validation before operational conclusions can be drawn.

Shift Analysis

Night shift shows the highest productivity (77.67), the longest resolution time, and the lowest command failure rate. Morning shift has the highest command failures, likely due to peak factory noise and concurrent device usage. These patterns were later tested statistically.

Platform Comparison

MS Teams outperforms Webex across every measured metric. Resolution efficiency is 33% higher on MS Teams. Productivity scores are 17% higher. Command success rates are 15% higher. Resolution time is approximately equal, which means MS Teams achieves better outcomes in the same time.

Device Health

All 20 RealWear devices fall in the medium battery risk category. No device has reached critical health status. However battery drain shows no correlation with command failures, indicating failures are driven by environmental and network factors rather than device health.

Noise Impact

High noise environments (above 85 dB) produce a command failure rate of 24.72% compared to 15.44% in low noise environments — a 60% increase. External microphone usage shows only a 0.40% improvement, making it practically ineffective as a noise mitigation tool.

Session Outcomes

92.91% of sessions completed successfully. 5% were dropped (technical failure) and 2% timed out (process failure). The 840 unsuccessful sessions represent significant operational risk at scale — even a small percentage translates to substantial downtime and unresolved issues.

Excel Dashboard

An interactive single-page Excel dashboard was built with KPI cards for total sessions, average productivity, issue resolution rate and downtime saved. Plant comparison, platform comparison, monthly trend and session status visuals were included with slicers for Plant, Shift and Platform filtering.

5. SQL Analytics

Advanced querying on SQLite database across 12 analytical categories

Technical Setup

All seven Excel sheets were imported into a SQLite database using a Python script with pandas and sqlite3. The database structure mirrors the star schema established during data modeling. A total of 54 queries were written covering everything from basic KPI aggregations to complex multi-table joins with window functions and CTEs.

Query Categories

Category	Queries	Techniques Used
KPI and Overview	7	COUNT, SUM, AVG, ROUND, CASE WHEN
Plant Performance	6	GROUP BY, RANK() OVER, CROSS JOIN, CTE
Worker Analysis	5	PARTITION BY, JOIN, PERCENT_RANK, NTILE
Device Analysis	4	JOIN, MIN/MAX normalization, Risk scoring
Platform Analysis	5	CASE WHEN, WHERE filter, PARTITION BY
Shift Analysis	4	GROUP BY, RANK, pattern identification
Noise Analysis	5	NTILE, LAG, CASE WHEN, threshold detection
Issue Type Analysis	4	GROUP BY, HAVING, AVG, ORDER BY
Time Series	5	LAG, rolling average, STRFTIME, trend
Advanced Window Functions	5	NTILE, cumulative SUM, ROW_NUMBER, PERCENT_RANK
Multi-table Joins	4	JOIN across 3+ tables, composite scoring
Business Intelligence	5	Classification logic, risk scoring, optimal combos

Selected Advanced Query Findings

Most productive session combination

MS Teams + 4G LTE + Night shift produces the highest average productivity score across all platform, connection and shift combinations — validated with PERCENT_RANK and minimum session threshold of 50 to control for low sample size bias.

Confounding variable detection

SQL analysis of external microphone vs command failures, stratified by noise category, revealed that the apparent negative effect of microphones is entirely explained by their deployment concentration in high noise environments — matching the statistical analysis findings.

Device risk classification

Min-max normalization applied to battery drain and command failure rate within SQL using window functions to create a composite risk score. All 20 devices classified as medium risk with no critical devices identified.

Plant department cross analysis

Maintenance department shows the highest resolution efficiency (46.89%) across plants while Production shows the lowest (45.15%). The 1.74% gap is too small for operational conclusions — flagged for statistical validation.

6. Statistical Analysis

Hypothesis testing, correlation analysis and outlier detection

Methodology

Eight statistical analyses were conducted using Python with scipy, statsmodels, pandas and numpy. All data failed Shapiro-Wilk normality tests. Log transformation was attempted but did not produce normality. Parametric tests were justified through Central Limit Theorem verification — every comparison group exceeded 2,300 samples, far above the minimum threshold of 30. Both p-values and effect sizes are reported throughout to distinguish statistical significance from practical significance.

Why Effect Size Matters

A p-value alone cannot guide business decisions. With 12,000 sessions even trivial differences produce statistically significant p-values. Effect size (Eta Squared for ANOVA, Cohen's d for T-Tests, Cramer's V for Chi-Square) was calculated alongside every test to measure whether differences are large enough to act on.

Descriptive Statistics — Key Findings

Metric	Mean	Std Dev	Skewness	CV%	Interpretation
Command Failures	5.60	4.58	1.15 right	81.84%	Most inconsistent — high variability
Resolution Time (min)	49.26	29.05	0.63	58.97%	High variability across sessions
Productivity Score	77.50	14.80	-0.52 left	19.09%	Stable — most workers above average
Command Success Rate	79.91	13.63	-0.53	17.06%	Most consistent metric
Noise Level (dB)	81.39	20.99	0.00	25.78%	Moderate variability

Hypothesis Test Results

Test 1 — Does noise level affect command failures?				REJECT H0
F-Statistic	P-Value	Eta Squared	Effect Size	
153.3946	< 0.001	0.0326	Small	

All three noise group pairs (High vs Low, High vs Medium, Low vs Medium) are significantly different. High noise adds 1.86 more failures per session versus low noise environments.

Finding and Recommendation

Noise significantly affects failures but explains only 3.26% of variance. Other factors drive the majority of failures. Noise reduction alone will not solve the command failure problem.

Test 2 — Is MS Teams significantly better than Webex?				REJECT H0
T-Statistic	T-Critical	Welch's DoF	Decision	
120.0369	1.6450	8,234.49	Strongly Reject	

Finding and Recommendation

T-Statistic of 120.04 versus critical value of 1.645 is one of the strongest findings in this analysis. MS Teams is not marginally better — it is overwhelmingly and consistently better across all 12,000 sessions.

Test 3 — Does shift timing affect productivity?

FAIL TO REJECT H0

F-Statistic	P-Value	Eta Squared	Effect Size
0.5126	0.5989	0.0001	Negligible

Finding and Recommendation

Excel showed night shift leading by 0.33 points. ANOVA confirms this is pure random variation. Shift timing explains 0.01% of productivity variance. This demonstrates why statistical validation must follow exploratory analysis.

Test 4 — Does external microphone reduce command failures?

FAIL TO REJECT H0

T-Statistic	Cohen's d	With Mic Mean	Without Mic
-13.8902	-0.2552	6.21 failures	5.05 failures

Finding and Recommendation

Confounding variable identified: noise level. Mic users show MORE failures because mics are deployed in high-noise zones where failures are inherently higher. This is a classic Simpson's Paradox scenario. Current microphone policy is based on a flawed assumption.

Test 5 — Does plant location affect productivity?

FAIL TO REJECT H0

F-Statistic	P-Value	Eta Squared	Note
2.0384	0.0862	0.0007	Borderline result

Bokaro 78.04 vs Vadodara 77.15 — gap of 0.89 points. P-value of 0.086 is borderline. A larger dataset may confirm significance. Monitor quarterly.

Finding and Recommendation

Unlike the shift test ($p = 0.599$), the plant test at $p = 0.086$ is close enough to warrant monitoring. Statistical power would increase with more data.

6 (cont.) — Correlation, Chi-Square and Outlier Detection

Correlation Analysis — What Drives Productivity?

Pearson correlation matrix computed across all 11 numeric variables. Ranked by absolute correlation with Productivity Score.

Rank	Factor	Correlation	Strength	Direction	Business Implication
1	Command Success Rate	+0.6996	Moderate	Positive	Primary productivity lever — optimize this first
2	Command Failures	-0.5805	Moderate	Negative	Most actionable metric for improvement
3	Noise Level dB	-0.1521	Negligible	Negative	Real but small — not primary focus
4	Downtime Saved	+0.0625	Negligible	Positive	Indirect relationship only
5–10	All others	< ±0.06	Negligible	Mixed	Battery, signal, duration — no meaningful impact

Core Productivity Insight

Voice command performance is the only factor with meaningful correlation to productivity. All operational factors including battery level, signal strength, session duration and resolution time have correlations below 0.07. Investment in anything other than voice command optimization will not move the productivity needle.

Chi-Square Test — Platform vs Plant Independence

Chi-Square = 3.0677	P-Value = 0.547	Cramer's V = 0.016	Result: Independent
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Platform choice is uniformly distributed — MS Teams 68%, Webex 32% — across all five plants. Platform deployment is centrally managed. This makes company-wide MS Teams standardization operationally straightforward with no plant-specific resistance to overcome.

Outlier Detection — IQR and Z-Score Methods

Column	IQR Outliers	IQR %	Z-Score Outliers	Z-Score %	Status
Command Failures	275	2.29%	94	0.78%	Investigate
Resolution Time	15	0.12%	1	0.01%	Monitor
Productivity Score	0	0.00%	0	0.00%	Clean
Noise Level dB	0	0.00%	0	0.00%	Clean
Battery Drain	0	0.00%	0	0.00%	Clean
Downtime Saved	0	0.00%	0	0.00%	Clean

IQR detected more outliers than Z-Score for Command Failures because the column is right skewed (skewness = 1.15). IQR is more sensitive to asymmetric distributions and is the more appropriate method for this column. The 275 sessions with above 17 command failures per session represent genuine operational anomalies — not natural variation.

7. Power BI Dashboard

Interactive business intelligence reporting

Dashboard Overview

Two interactive Power BI dashboards were built connecting directly to the Excel dataset. The dashboards enable real-time filtering by Plant, Shift and Platform through slicers, allowing management to drill into specific operational contexts.

Dashboard	Key Visuals	Purpose
Executive Summary	KPI cards (Sessions, Productivity, Resolution Rate, Downtime)	Top-level productivity overview, Session status and counts
Plant and Operations	Shift x Plant stacked chart, Plant scorecard table with all KPIs	Operational deep dive, Departmental management

Key DAX Measures

Custom DAX measures were written for Issue Resolution Rate (percentage of sessions where Issue_Resolved equals Yes), Downtime Saved in Hours (sum of minutes divided by 60), Average Productivity Score, and Session Completion Rate. These measures power all KPI cards and enable dynamic recalculation when slicers are applied.

8. Business Recommendations

Evidence-based actions prioritized by impact and feasibility

IMMEDIATE

1. Standardize MS Teams across all five plants

Evidence: T = 120.04, 17% higher productivity, 33% better resolution efficiency

Decommission Webex for RealWear sessions. Update device configurations and IT infrastructure to support 100% MS Teams deployment. No plant-specific change management required — platform choice is centrally managed and uniform.

Expected impact: Every session currently on Webex is operating at measurable productivity disadvantage. Full migration eliminates this gap across all 3,811 annual Webex sessions.

IMMEDIATE

2. Investigate the 275 high-failure anomalous sessions

Evidence: IQR outlier detection — 275 sessions with above 17 command failures

Isolate the 275 sessions and perform root cause analysis. Identify common characteristics across plant, device model, shift, connection type and noise level. These sessions likely share a pattern that when identified can be prevented through targeted maintenance or protocol changes.

Expected impact: These sessions skew operational benchmarks and represent genuine device or environmental failures that should not recur.

SHORT TERM

3. Focus voice command optimization efforts

Evidence: Command Success Rate $r = +0.70$, Command Failures $r = -0.58$ with Productivity

Invest in RealWear device firmware updates for speech recognition improvement. Implement worker training on effective voice command usage — slower and clearer pronunciation with simpler shorter commands. This is low cost and immediately actionable.

Expected impact: These are the only two factors with meaningful correlation to productivity. Every other operational factor has negligible impact.

SHORT TERM

4. Revise external microphone deployment policy

Evidence: Confounding variable identified — noise level drives both mic usage and failures

Do not procure additional external microphones as a noise solution. Instead, evaluate microphone effectiveness within matched noise environments. Investigate noise cancellation and beamforming microphone technology for high-noise zones. Consider restricting device usage to low and medium noise areas where practical.

Expected impact: Current policy is based on a flawed assumption. Resources directed at microphone procurement will not reduce command failures.

ONGOING

5. Monitor plant-level performance quarterly

Evidence: Borderline ANOVA $p = 0.086$ — not yet significant but worth tracking

Track plant productivity monthly in Power BI dashboard. Retest with expanded dataset (targeting 25,000+ sessions) to determine whether Bokaro Manufacturing Hub advantage over Vadodara Chemical Unit is statistically confirmable.

Expected impact: *Borderline significance suggests weak underlying differences. Early detection of genuine plant divergence enables proactive intervention.*

ONGOING

6. Implement predictive device maintenance schedule

Evidence: All 20 devices at medium battery risk — none critical but none healthy

Schedule preventive maintenance for all 20 devices before any reaches critical health status. Establish a health score threshold (below 50) as an automatic maintenance trigger. Track battery drain trends monthly.

Expected impact: *Prevents reactive maintenance costs and session disruptions from unexpected device failures.*

9. Conclusions and Next Steps

Summary of analytical outcomes and future work

What This Analysis Proved

Platform choice is the single most impactful controllable variable.

MS Teams delivers statistically and practically superior results with overwhelming evidence. This is actionable immediately without further analysis.

Voice command performance is the only meaningful driver of productivity.

Correlation analysis confirms that command success rate and failures explain the majority of productivity variation. Operational factors like battery, signal and session duration are irrelevant.

Statistical validation overturned several Excel-based assumptions.

Night shift productivity advantage, external microphone effectiveness, and plant-level performance differences all appeared significant in Excel but were statistically confirmed to be either negligible or confounded.

Data quality is strong with targeted anomalies.

97.71% of sessions are clean. The 275 anomalous sessions are concentrated in Command Failures and represent specific operational incidents rather than systemic data quality issues.

The borderline plant result warrants continued monitoring.

Unlike shifts ($p = 0.599$), the plant test at $p = 0.086$ sits close enough to significance that expanding the dataset could reveal genuine performance differences.

Future Scope

- Machine learning models to predict command failures before sessions begin using pre-session variables (plant, device, noise category, connection type, shift)
- Time-series forecasting of monthly session volume to support resource planning and device allocation across plants
- Controlled experiment design to properly test external microphone effectiveness within matched noise environments — removing the confound identified in this analysis
- Worker performance clustering using K-Means to segment workers into performance groups and identify targeted training needs
- Real-time Power BI dashboard connected to live data for continuous operational monitoring rather than retrospective analysis
- Expand dataset to 50,000+ sessions over 24 months to increase statistical power and definitively resolve the borderline plant performance question

RealWear Industrial Operations Analytics and Smart Factory Intelligence System

Prepared by Dev Pandey · Data Analyst · January 2025 – January 2026

Tools: Microsoft Excel · SQL (SQLite) · Power BI · Python (scipy, pandas, statsmodels)